Xochitl Ortega

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Education

University of California, Berkeley

Master of Science, **Mechanical Engineering** *GPA:3.87* **May 2026**

Bachelor of Science, **Mechanical Engineering** *GPA:3.34* **May 2025**

UC Berkeley Certificate in Design Innovation

Relevant Coursework: Computational Biomechanics Across Multiple Scale, Human-Centered Design Methods, Embedded Microprocessor-Based Systems, Adv Designing for the Human Body, Design of Morphing Materials, Intro to Product Dev

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Skills

Design & Development: Design Controls, Design of Experiments (DOE), Design for Manufacturing (DFM), Human-Centered Design, Design Thinking, Rapid & Iterative Prototyping

CAD & Simulation: SolidWorks, Fusion 360, AutoCAD, Onshape, ANSYS (FEA), GD&T,

Fabrication & Manufacturing: CNC Milling, Lathe, Machine Shop, 3D Printing, Injection Molding, Laser Cutting, Waterjet Cutting, Soldering, Tooling Design, Soft Goods & Textile Construction, Line Optimization

Testing & Characterization: Instron Tensile Testing, X-Ray Diffraction (XRD), Root Cause Analysis, Failure Analysis, Embedded Sensing, Data Acquisition

Software: MATLAB, Simulink, Python, LabVIEW, Miro, Figma, MS Office (Excel, Word, PowerPoint)

Research

Wearables Research Assistant **June – Aug 2026**

Aalto University, Dept. of Computer Science | Espoo, FI

- Translating movement and physiological rhythms into wearable, everyday-usable form by iterating across textiles, soft electronics, and embedded sensing under Professor Deepika Yadav
- Researching how fatigue and exhaustion surface across menstrual life stages, using somatic body-journaling data to map the connection between women's self-knowledge and their measured physiological signals

Prototype Research Assistant **Feb 2026 - May 2026**

Morphing Matters Lab, Dept. of Mech E | Berkeley, CA

- Engineered self-actuating textile "e-sequins" by integrating embedded permanent magnets, 3D-printed TPU, and machine-sewn assembly into a functional wearable system
- Advanced sequin actuation from binary manual flips to continuous, controllable motion; the first demonstration of its kind under Professor Lining Yao
- Designed and 3D-printed custom test fixtures to isolate actuation variables, and built driving electronics (ESP32 microcontroller, MOSFET switching, rotary-encoder GUI) to tune pulse intensity, duration, and frequency

Materials Synthesis Research Assistant **June - Aug 2025**

Nanyang Technological University | Singapore, SG

- Selected as 1 of 56 Global Connect Fellows from 4,500+ applicants across 25+ countries (~1% acceptance) for NTU's fully funded research fellowship
- Synthesized multi-metallic FeCoZnNa nanoparticle catalysts for CO_2 conversion using Direct Joule Synthesis (DJS) by translating machine-learning-predicted compositions into physical samples through precursor weighing, ball milling, and vacuum drying
- Optimized synthesis parameters, pulse duration, ramp rate, and temperature, via real-time infrared feedback to reduce contamination and processing time relative to conventional furnace methods
- Characterized crystal structure and phase purity by X-ray diffraction (XRD), confirming homogeneous formation and batch-to-batch reproducibility across synthesis run

Research Assistant **Aug 2024 - Jan 2026**

Berkeley Biomechanics Lab, Dept. of Mech E | Berkeley, CA

- Investigated mechanical degradation of lab-aged polyurethane (PE-PUR) foam used in BiPAP/CPAP devices to assess long-term material safety, motivated by prior device recalls attributed to foam breakdown
- Identified a ~33% reduction in elastic modulus under high-humidity hydrolytic degradation via a custom-designed DOE simulating real-world environmental stressors
- Performed Instron tensile testing, density, and visual property measurements to benchmark material failure thresholds, and authored a technical report supporting long-term safety evaluation

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Employment

Mechanical Engineer Intern **June - Aug 2024**

L'Oréal | Little Rock, AR

- Cut part cost 96%, eliminating a supply chain dependency and slashed lead time from 9 weeks to 24 hours by bringing torque chuck rubber production in-house and designing replacement molds
- Ran design of experiments (DOE) on lipstick bullet manufacturing parameters, establishing standardized baseline recipes by color category and integrating pre-calculated parameters into line control systems to improve operator efficiency and batch consistency
- Identified push-out pressure as a previously undocumented critical failure factor in matte formulations, resolving drop-test failures across six lots (163K units, $204K in buybacks, 4-week release delays)

Design Engineer Intern **June - Aug 2023**

Variantz | Singapore, SG

- Supported Singapore's smart home technology market through expo research and competitive analysis to support the company's expansion from B2B/B2G into consumer products, identifying unmet needs in connected household devices
- Developed two IoT product concepts in SolidWorks, a filtering smart showerhead with water usage tracking and a canine health and activity tag, integrating embedded sensing with a companion app and designing customizable exteriors and replaceable filter modules for aftermarket resale

SMASH Rising Intern **June - July 2022**

Zillow | Remote

- Designed and prototyped a gamified app concept in Figma to guide first-time buyers through the home purchase process, iterating on the flow from focus group feedback and integrating in-app connections to Zillow agents

SMASH Rising Intern **June - July 2021**

AutoDesk | Remote

- Designed "FRESH," a Figma-based website, leveraging in-depth research on API contracts and cloud licensing to optimize the use of AutoDesk's resources for clearer understanding and improved utilization of licensing information
- Improved error resolution for customers by 27% through multiple rounds of ideation, ensuring contract components could be effectively coded, tested, and validated

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Campus Employment

Course Staff **June - Aug 2026**

Dynamic Systems and Feedback, College of Engineering **(Remote)**

- Supporting instruction, grading, and exam administration for a 70+ student mechanical engineering course

Student Assistant & Advisor **Aug 2025 - May 2026**

Engineering Student Services, College of Engineering

- Advised undergraduate students with academic planning, research opportunities, and graduate school applications
- Managed front desk operations, advisor scheduling, and support for College of Engineering events

Course Staff **Aug 2025 - Jan 2026**

Design & Creativity Lab, Haas School of Business

- Supported hands-on instruction, grading, and attendance as the sole TA/reader for a 55-student design course

Student Assistant **Jan 2023 - May 2025**

Center for Access to Engineering Excellence, College of Engineering

- Managed a student lending library and study center operations, sourcing supplies cost-effectively and coordinating academic resources for engineering students
- Hosted academic events and career workshops, and oversaw student competition clubs, ensuring compliance with university standards

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Projects

Design of Microprocessor-Based Mechanical Systems - ME235 | "Robogotchi" **Spring 2026**

- Designed a Tamogothchi inspired 3D-printed differential-drive pet robot chassis integrating 8+ electromechanical subsystems around a single microcontroller, balancing mechanical constraints (wheel clearance, sliding access panel, wiring routing) with real-time performance requirements

Human–AI Design Methods - ME120 | "Stride Recovery" **Spring 2026**

- Designed a wireless, smart Rehabilitation sleeve Integrating electromyography (EMG) sensing to deliver real-time tracking of muscle activation, fatigue, and bilateral hamstring symmetry
- Prototyped a companion smartwatch interface via protoboarding to establish wireless data synchronization, enabling real-time biofeedback display and signal validation metrics
- Leveraged bilateral symmetry analysis and real-time data feedback to build an evidence-based system aimed at reducing post-injury guesswork and Athletic reinjury rates

Critical Systems of Development - DevEng202 | **Spring 2026**

"Medical Device Equity in Low-Resource Analysis"

- Analyzed systemic barriers to medical device access in rural and Indigenous Mexico
- Mapped stakeholder power and lifecycle failures across procurement, maintenance, and post-market surveillance to identify where device design must account for local infrastructure and repair capacity

Computational Biomechanics Across Multiple Scales - ME120 | **Spring 2026**

"Biofilm Mechanics Blocking Tissue Repair"

- Developed a multiscale biomechanical simulation framework to analyze tensile deformation and stress fields in bacterial biofilms
- Mapped polymer network Dynamics to a 2D continuum mesh to evaluate von Mises stress under wound contraction boundary conditions

Morphing Matters - ME292C | "Auto-Lifting Dress" **Fall 2025**

- Led a team of four to develop an adaptive morphing wearable that responds to user acceleration and elevation
- Explored materials and fabrication techniques through iteration to optimize functionality, durability, and design
- Created a wireless protoboard control to manually control different portions of the dress

Design, Evaluate, and Scale Development Technologies - DevEngC200 | "BahaWatch" **Fall 2025**

- Partnered with a Philippines-based disaster resilience nonprofit to evaluate low-cost wireless flood sensor technologies and identify infrastructure gaps in early-warning communication

Advanced Designing for the Human Body - ME278 | "Hand Tremor Aid" **Fall 2025**

- Applied human-centered design and engineering methods to analyze tremor-related motor impairments and prototype personalized assistive devices that enhance fine motor control and stability

Mechatronics Design - ME102B | "EchoHand" **Spring 2025**

- Improved dexterity across 17 DOF through the design of a 3D-printed prosthetic hand in SolidWorks
- Integrated MediaPipe hand tracking to replicate natural hand movements for users with muscle atrophy

Experimentations And measurements - ME103 | "Analysis of SpaceX Starship Model" **Fall 2024**

- Conducted controlled testing of angles of attack and velocities of the 1:200 scale of SpaceX Starship Rocket using the wind tunnel

The Science and Engineering of Cooking - ME126 | **Fall 2024**

"TempDrop - A New Cooking Apparatus

- Designed and fabricated an automated boiling system using ESP32 and thermocouple technology to monitor water temperature
- Enabled hands-free cooking by releasing ingredients at optimal boiling points and notifying users via smartphone alerts

Innovation and Design Thinking in Business - UGBA190D | "EcoEats" **Fall 2024**

- Developed an app and curriculum focused on food sustainability,
- Provided educational resources on healthy eating, sustainable purchasing options, and community engagement through recipe sharing, while heavily focusing on the design process

Introduction to Product Development - ME110 | "MotionGlow" **Summer 2024**

- Designed an outlet cover equipped with motion and light sensors, automatically activating LED lights for improved visibility when plugging in devices in dark environments
- CADded using SolidWorks and performed thermal analysis to assess product durability under high temperatures with simulations showing heat impact around outlet holes for realistic performance testing

Electronics for the Internet of Things - ME100 | "Iot Cup Pong" **Fall 2023**

- Automated cup pong using microcontrollers & laser sensors to track user scores reliably,
- developed an accurate hardware-software interface,
- utilized an MQTT wireless network for real-time data exchange & control
- Designed and assembled circuit schematics with various components, ensuring functionality and preventing component overload

Statistics and Data Science for Engineers - E178 | "AI/ML Star Predictor" **Fall 2023**

- Achieved 100% accuracy in star classification using K-Nearest Neighbors and neural networks n python on a 400+ star NASA dataset

RoboBears - ME198 | Team Lead **Fall 2023**

- Designed and CADed an angled wedge with a rotating attacker for a 1-pound competitive robot, focusing on optimizing functionality and durability during the manufacturing process

Manufacturing and Design Communication - ME29 | "Lecture Hall Desk Space Optimizer" **Fall 2022**

- Designed a screw-on desk extender for lecture halls to optimize space by attaching a water bottle holder, space for a portable charger, and a pencil holder
- Assembled parts on SolidWorks and the physical prototype, compiled project documentation, and mitigated communication challenges within the group, to develop and showcase a desk extender to maximize space

Three-Dimensional Modeling for Design - ME26 | "Wind Turbine" **Spring 2022**

- Developed, 3D printed, and tested a wind turbine to optimize speed and load-to-deflection ratio, achieving 1 7 watts and ranking second in class for efficiency in load-to-deflection performance
- Actively drove team collaboration, conducted FEA structural analysis, and executed fabrication and assembly

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Leadership

Hispanic Engineers & Scientists (SHPE) | *Mentor*, UC Berkeley **Aug 2024 - May 2025**

- Mentored students by providing guidance on academic planning and professional development, supporting goal setting and skill-building
- Managed website updates and maintenance as Webmaster, improving organization, accessibility, and user experience

LeaderShape Institute | *On-Site Student Coordinator*, UC Berkeley CoE **Jan 2024**

- Facilitated a week-long leadership program for 60 engineering students, fostering communication, team-building, and inclusive collaboration to develop leadership skills and empower diverse voices

Berkeley Mentorship Cohort | *Mentor* **,** UC Berkeley CoE **Aug 2023- May 2024**

- Awarded Best Mentor for guiding an incoming freshman through academic planning, study strategies, campus resources, and the CoE transition

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Community

Gradbridge (purdue/ berkeley) | Participant **2025**

Hispanic Engineers & Scientists (SHPE Region 1) | Webmaster + *Mentor***,** UC Berkeley **2021 - 2025**

Apple Pathways Alliance | *Participant* **2023**

Apple Women in Science and Engineering Mentorship | *Participant* **2022**

Raytheon Technologies Engineering Scholarship | *Recipient* **2021-2025**

SMASH Rising | *SMASH Intern* **2021-2022**

Girls Inc. (youth)

SMASH Academy (youth)